P-A-450B
July 1, 1975
SUPERSEDING
Fed. Spec. P-A-450A
March 6, 1964

# FEDERAL SPECIFICATION

## ALKALI, LAUNDRY, CONTAINING CARBOXYMETHYLCELLULOSE

**This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.**

## 1. SCOPE

1.1 Scope. This specification covers one type and one grade of laundry alkali (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

| | | |
|---|---|---|
| L-P-378 | - | Plastic Sheet and Strip, Thin Gauge, Polyolefin |
| O-S-571 | - | Sodium Carbonate Anhydrous, Technical |
| O-S-598 | - | Sodium Hydroxide, Technical |
| RR-S-366 | - | Sieve, Test |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |
| PPP-D-705 | - | Drum: Metal Shipping, Steel, (Over 12 and Under 55 Gallon) |
| PPP-D-723 | - | Drums, Fiber |

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 536 - Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 7930

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

16 CFR 1500 - Federal Hazardous Substances Act

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

D 501 - Sampling and Chemical Analysis of Alkaline Detergents
D 1172 - Test for pH of Aqueous Solutions of Soaps and Detergents

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Sodium hydroxide. The sodium hydroxide shall conform to O-S-598.

3.1.2 Sodium carbonate. The sodium carbonate shall conform to O-S-571.

3.1.3 Sodium carboxymethylcellulose. The sodium carboxymethylcellulose shall be tested as specified in 4.4.3.

3.2 Composition. The laundry alkali shall consist of a combination of sodium hydroxide, sodium carbonate, and sodium carboxymethylcellulose. The composition of the laundry alkali shall conform to the composition shown in table I, when tested as specified in table III.

TABLE I. Composition

| Ingredient | Percent by weight (minimum) |
|---|---|
| Sodium hydroxide | 27.0 |
| Sodium carbonate | 62.0 |
| Sodium carboxymethylcellulose (active) | 2.5 |

3.3 Characteristics of laundry alkali. The laundry alkali shall conform to the characteristics shown in table II, when tested as specified in table III.

3.4 Form. The laundry alkali shall be in powdered or granular form and shall conform to the fineness requirements specified in table II when tested as specified in table III.



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TABLE II. Physical and chemical requirements

| Characteristics | |
|---|---|
| Total alkalinity as $Na_2O$, minimum, percent | 59.0 |
| pH value of a 0.5 percent aqueous solution at 25 ± 0.5°C. (minimum) | 12.2 |
| Fineness of granulation: | |
| Maximum retained on 10 mesh sieve, percent | 6.0 |
| Maximum passing 200 mesh sieve, percent | 27.0 |
| Moisture, maximum, percent | 4.0 |
| Matter insoluble in water, maximum, percent | 0.1 |

3.5 Workmanship. The finished product shall be clean, uniform, free flowing, and shall not lump or cake.

3.6 Labeling. Each shipping container shall be labeled in accordance with the Federal Hazardous Substances Act.

3.6.1 Danger label.

3.6.1.1 All shipping containers shall be labeled with the following warning information:

DANGER: MAY CAUSE BURNS TO SKIN AND EYES.
HARMMUL IF SWALLOWED.

Contains ALKALI.
Avoid contact with skin, eyes; and mucous membranes.
In case of external contact, flush with water.
For eyes, flush with water for 15 minutes and get immediate medical attention.
If swallowed, give large quantities of water or milk. Call physician immediately.
Keep out of the reach of children.

3.6.2 Caution label.

3.6.2.1 All shipping containers shall be labeled with the following caution information:

CAUTION: CONCENTRATED ALKALI -- KEEP COVER CLOSED, WEAR GOGGLES OR FACE SHIELD, RUBBER GLOVES AND PROTECTIVE CLOTHING WHEN HANDLING



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.2.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.3 Inspection of the end item. The end item shall be examined for the defects in the applicable sub-paragraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.2.3.4. Random samples shall be drawn from each lot of end items offered for inspection. The lot size for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled containers of the specified capacity for examination in 4.2.3.1, 4.2.3.2, and 4.2.3.3.

4.2.3.1 Examination of the primary container for visual defects in construction and markings. The sample unit for this examination shall be one filled container.

| Examine | Defect |
|---|---|
| Construction | Not in container specified.<br>Any leakage or sifting of contents.<br>Any split, break, dent, hole, tear, puncture, or other defect which may affect serviceability.<br>Seams or closures not as specified.<br>Polyethylene liner missing or not sealed when specified. |



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| Examine | Defect |
|---|---|
| Markings (label or printing) | Omitted, illegible, incorrect, incomplete, not in accordance with requirements.<br>Label or printing not clear, smeared. |

4.2.3.2 Examination for defects in net contents. The sample unit for this examination shall be one filled container. The lot shall be unacceptable if the average net content per container is less than the specified or indicated quantity.

4.2.3.3 Examination of laundry alkali for defects in workmanship. The sample unit for this examination shall be the contents of one filled container.

| Examine | Defect |
|---|---|
| Workmanship | Not in powdered or granular form<br>Not clean; not uniform; any foreign matter.<br>Not free flowing.<br>Lumping or caking |

4.2.3.4 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per hundred units, shall be as follows:

| Examination paragraph | Inspection levels | AQL's |
|---|---|---|
| 4.2.3.1 | I | 2.5 |
| 4.2.3.2 | S-2 | --- |
| 4.2.3.3 | S-3 | 1.5 |

4.3 Testing of the end item. The laundry alkali shall be tested for the characteristics listed in tables I and II in accordance with test methods specified in table III for each lot presented for examination. The sample specimen shall be one pound of laundry alkali. The sample size shall be five. Each sample specimen shall be taken from a different shipping container of the lot. The test sample shall consist of a two pound composite of the five sample specimens. Unless otherwise specified in the contract or order, duplicate determinations shall be performed on the composite. For lot acceptability, the average of the two determinations shall comply with the specified requirements for each characteristic.



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4.4 Tests.

4.4.1 Sodium hydroxide. The sodium hydroxide shall be determined in accordance with ASTM Standard D 501, sections 8, 9, and 10.

4.4.2 Sodium carbonate. The sodium carbonate shall be determined in accordance with ASTM Standard D 501, section 11.

4.4.3 Sodium carboxymethylcellulose (CMC).

4.4.3.1 Apparatus. The apparatus required shall be available in the laboratory.

TABLE III. Instructions for testing end item

| Characteristic | Specification reference | | Results reported as |
|---|---|---|---|
| | Requirement | Test paragraph | Numerically to nearest 1/ |
| Sodium hydroxide | Table I | 4.4.1 | 0.1 percent |
| Sodium carbonate | Table I | 4.4.2 | 0.1 percent |
| Sodium carboxymethyl-cellulose (active) | Table I | 4.4.3 | 0.1 percent |
| Total alkalinity as $Na_2O$ | Table II | 4.4.4 | 0.1 percent |
| pH value | Table II | 4.4.5 | 0.1 pH unit |
| Particle size: Maximum retained on 10 mesh sieve | Table II | 2/ | 0.1 percent |
| Maximum passing 200 mesh sieve | Table II | 2/ | 0.1 percent |
| Moisture | Table II | 4.4.6 | 0.1 percent |
| Matter insoluble in water | Table II | 4.4.7 | 0.1 percent |

1/ Test reports shall include all values on which results are based.

2/ Method 2101 of FED-STD-536.

4.4.3.2 Reagents. Where applicable, analytical reagent grade chemicals and distilled water shall be used throughout the tests.

a. Dilute hydrochloric acid (3N).
b. Potassium dichromate solution (approximately 0.1N).
c. Sodium thiosulfate solution (0.1N) Standardized.



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d. Starch indicator.
e. Diatomaceous earth (free from organic matter).
f. Potassium iodide (free from iodate).

4.4.3.3 Procedure. Weigh to the nearest milligram a 1 g sample of laundry alkali into a 250 ml centrifuge bottle standing in a suitable size beaker. Add 90 ml of methyl alcohol and 10 ml of dilute hydrochloric acid. Heat to boiling on a water bath and simmer gently for 15 to 20 minutes. Cool and add 1 g of diatomaceous earth and stir the mixture. Centrifuge. Decant the supernatant liquid and wash the residue twice with 50-ml portions of methyl alcohol, centrifuging after each washing. Finally decant and dry the residue, cautiously at first to prevent spattering, for 2 hours at 105°C. Transfer the residue from the centrifuge bottle to a 1-liter g.s. I flask with three 10-ml portions of water. Add exactly 50 ml of potassium dichromate solution and 60 ml of concentrated sulfuric acid. Maintain the temperature of the mixture between 80 to 90°C for 30 minutes. Dilute to 400 to 500 ml with water. Cool to room temperature and add 2 g of potassium iodide. Stopper the flask and swirl to mix, and allow to stand in the dark for 10 minutes. Rinse the stopper and the inner bulb of the flask with water and titrate the liberated iodine with sodium thiosulfate solution (0.1N) until the solution is only faint yellow in color. Add starch indicator and continue the titration to the discharge of the blue color. Carry out a blank omitting only the sample. Calculate the percent sodium carboxymethylcellulose as follows:

$$\frac{[\text{ml sodium thiosulfate (blank)} - \text{ml sodium thiosulfate (sample)}] \times N \times 0.8}{\text{wt. of sample (g)}}$$

where: N = Normality of sodium thiosulfate solution.

4.4.4 Total alkalinity as $Na_2O$. The total alkalinity shall be determined in accordance with ASTM Standard D 501, section 4, except a 5 gram sample, weighed to the nearest milligram shall be used.

4.4.5 pH value. The pH value shall be determined in accordance with ASTM Standard D 1172, section 4, except a 5 gram sample weighed to the nearest milligram shall be used.

4.4.6 Moisture. Weight a 2 gram sample to the nearest milligram in a tared, glass-stoppered weighing bottle about 6 to 8 centimeters in diameter and about 2 to 4 centimeters in depth, and dry to constant weight in an air oven at a temperature of 105° ± 2°C. (Constant weight is considered attained when successive heating for 1-hour periods shows a loss (or gain) of not more than 0.1 percent). Loss in weight will be considered as moisture. Calculate the percent moisture as follows:



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$$\text{Percent moisture} = \frac{\text{Loss in weight X 100}}{\text{Weight of original sample}}$$

4.4.7 Matter insoluble in water. The matter insoluble in water shall be determined in accordance with ASTM Standard D 501, sections 28 and 29.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One hundred pounds of laundry alkali shall be packed in a metal drum conforming to type III of PPP-D-705, or in a fiber drum of appropriate size conforming to type II, grade A of PPP-D-723. The interior of each drum shall be lined with a minimum 0.003-inch thick polyethylene bag fabricated from material conforming to type I, grade A, finish 1 of L-P-378. Each bag shall be closed by heat sealing. Prior to or during the final closure operation, excess air within the bag shall be expelled. Alternatively, the bag closure shall be effected by securing the open end with two double-looped wire ties. The first wire tie shall be applied as close to the contents as possible. The second wire tie shall be applied approximately one inch from the first wire tie, with the twisted ends positioned opposite to the ends of the first wire tie. The ties shall be paper or plastic coated soft wire.

5.1.1.2 Twenty-five pounds of laundry alkali shall be packed in a fiberboard shipping container conforming to style FTC, grade V2s of PPP-B-636. Each container shall be provided with a liner as specified in 5.1.1. Each container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.1.2 Level B.

5.1.2.1 One hundred pounds of laundry alkali shall be packed in a metal drum as specified in 5.1.1.1, or in a fiber drum of appropriate size conforming to type I, grade A of PPP-D-723. Each drum shall be provided with a liner as specified in 5.1.1.1.

5.1.2.2 Twenty-five pounds of laundry alkali shall be packed as specified in 5.1.1.2, except the container shall be grade V3c, V3s, or V4s fiberboard material.

5.1.3 Level C. The laundry alkali shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.



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5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

5.2.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The laundry alkali, containing carboxymethylcellulose, covered by this specification, is intended for use with synthetic detergents or soaps, as an alkali, in the heavy duty laundering of cotton (white and colored) fabrics. It is suitable for use in soft or hard water. It is not intended for use in laundering woolen, silk or synthetic fabrics.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement doucments:

(a) Title, number, and date of this specification.
(b) Quantity and level of packing (see 5.1).
(c) Marking requirements (see 5.2).
(d) Size of container required (see 5.1).
(e) Selection of applicable levels of packaging and packing (see 5.1).

Custodians:

Army - GL
Air Force - 84

Review activities:

Army - MD

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS
HEW - FDA

Project No. 7930-0314

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